

OLA Ride Booking Analysis

1. Project Overview

This project analyzes OLA ride-booking data from 100,000 bookings to understand booking patterns, cancellations, ride performance, customer experience, and revenue. The goal is to uncover operational trends and identify factors affecting successful rides, cancellations, ratings, and booking value to support better business decisions.

2. Dataset Summary

- Rows: 100,000
- Columns: 20
- Key Features:
 - Booking details: Date, Time, Booking ID, Booking Status, Customer ID
 - Ride details: Vehicle Type, Pickup Location, Drop Location, Ride Distance
 - Performance metrics: Average V TAT, Average C TAT, Driver Ratings, Customer Rating
 - Cancellation details: Customer/Driver cancellations and cancellation reasons
 - Revenue & ride status: Booking Value, Incomplete Rides, and Incomplete Ride Reasons
- Missing Data: Missing values are present in TAT, cancellation reason, incomplete ride reason, and rating columns.

3. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MYSQL to answer key business questions:

1. Successful Bookings – Analyzed the total number of successfully completed rides to measure overall booking performance.

id	Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location
	25-07-2026	10:20:00 PM	CNR2940424040	Success	CID225428	Bike	Coimbatore	Tiruppur
	30-07-2026	07:59:00 PM	CNR2982357879	Success	CID270156	Prime SUV	Madurai	Dindigul
	02-07-2026	09:02:00 AM	CNR1797421769	Success	CID939555	Mini	Tiruchirappalli	Thanjavur
	13-07-2026	04:42:00 AM	CNR8787177882	Success	CID802429	Mini	Erode	Bhavani
	23-07-2026	09:51:00 AM	CNR3612067560	Success	CID476071	Bike	Vellore	Ranipet
	29-07-2026	11:33:00 PM	CNR4787583516	Success	CID923404	Prime Plus	Karur	Namakkal
	26-07-2026	04:03:00 AM	CNR7943634301	Success	CID647026	Prime Plus	Cuddalore	Puducherry
	27-07-2026	01:18:00 PM	CNR4524472111	Success	CID540929	Auto	Villupuram	Tindivanam
	16-07-2026	09:54:00 AM	CNR8181602032	Success	CID167642	Bike	Sivakasi	Virudhunagar
	02-07-2026	10:25:00 AM	CNR8090918544	Success	CID640151	Bike	Ramanathapuram	Paramakudi

2. Ride Distance by Vehicle Type – Compared the average ride distance across different vehicle types to understand travel patterns and vehicle usage.

Vehicle_Type	avg_distance
Prime Sedan	15.7649
Bike	15.5331
Prime SUV	15.2745
eBike	15.5806
Mini	15.5101
Prime Plus	15.4475
Auto	6.2381

3. Total Customer Cancellations – Analyzed the number of rides cancelled by customers to identify cancellation patterns and potential service issues.

COUNT(*)
10499

4. Top 5 Customers – Identified the top five customers based on their total booking value to understand high-value customer contributions.

Customer_ID	total_rides
CID954071	5
CID539191	4
CID189965	4
CID268274	4
CID952434	4

5. Total Driver Cancellations – Analyzed rides cancelled by drivers due to personal and vehicle-related issues to identify operational challenges affecting ride completion.

COUNT(*)
6542

6. Maximum & Minimum Driver Rating – Identified the highest and lowest driver ratings to evaluate overall driver performance and service quality.

max_rating	min_rating
5	3

7. UPI Payments – Analyzed the number of rides paid through UPI to understand customer payment preferences and digital payment adoption.

Booking_Value	Payment_Method	Ride_Distance
386	UPI	40
173	UPI	41
330	UPI	8
378	UPI	18
343	UPI	23
343	UPI	17
286	UPI	44
141	UPI	35
214	UPI	2
461	UPI	49
1102	UPI	41
444	UPI	6

8. Average Customer Rating by Vehicle Type – Compared the average customer rating across different vehicle types to evaluate service quality and customer satisfaction.

Vehicle_Type	avg_customer_rating
Prime Sedan	4.001588655506982
Bike	3.993376395883525
Prime SUV	3.999377501111586
eBike	3.98785403050109
Mini	3.9977312970341075
Prime Plus	4.009498622589555
Auto	3.998810952329009

9. Total Successful Ride Value – Calculated the total booking value generated from successfully completed rides to measure overall ride revenue.

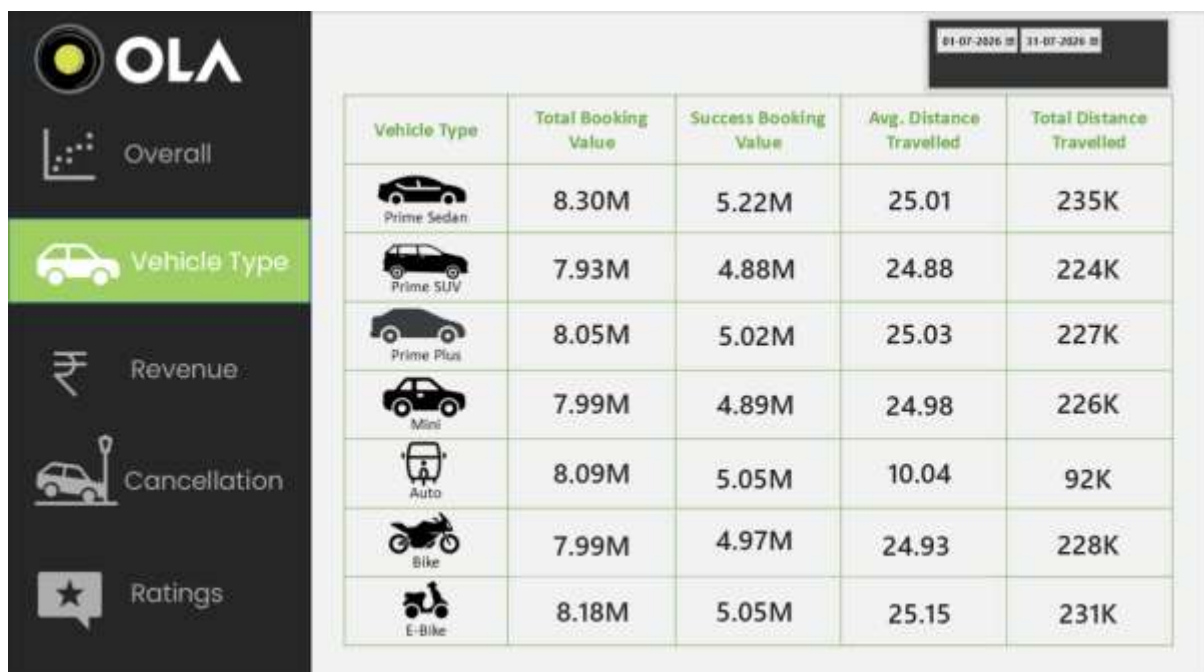
total_successful_ride_value
35080467

10. Incomplete Ride Reasons – Analyzed the reasons for incomplete rides to identify operational issues and factors affecting successful ride completion.

Booking_ID	Incomplete_Rides_Reason
CNR5176704322	Customer Demand
CNR9312632867	Vehicle Breakdown
CNR7924302885	Customer Demand
CNR1640228587	Other Issue
CNR7623690602	Other Issue
CNR9590311980	Customer Demand
CNR5863244684	Customer Demand
CNR9526078867	Customer Demand
CNR7154043084	Customer Demand
CNR3193710797	Other Issue
CNR7073850950	Customer Demand
CNR9952584604	Customer Demand
CNR5433575259	Vehicle Breakdown
CNR3575066041	Vehicle Breakdown
CNR7537935962	Customer Demand

4. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.







5. Business Recommendations

- **Reduce Cancellations** – Identify major customer and driver cancellation reasons and take corrective actions.
- **Improve Ride Completion** – Address common incomplete-ride issues to increase successful bookings.
- **Optimize Vehicle Performance** – Focus on vehicle types with strong demand, ratings, and ride performance.
- **Improve Customer Experience** – Use customer ratings and waiting times to identify and improve service quality.
- **Optimize Revenue** – Focus on high-value rides, popular locations, and vehicle types to improve overall booking revenue.